EARTHQUAKE SEQUENCE

- **Foreshocks (The Warning Tremors)**
  - These are smaller earthquakes that happen **before** the biggest shake in the same area.
  - They occur because tectonic stress starts building up underground, causing small sections of rock to crack or slip first.
  - *Note:* Not all major earthquakes have noticeable foreshocks.
- **Mainshock (The Big Event)**
  - This is the **largest and most powerful** earthquake in the sequence.
  - It releases the vast majority of the built-up energy along the fault line and causes the most severe shaking and potential damage.
- **Aftershocks (The Settling Tremors)**
  - These are smaller earthquakes that happen **after** the mainshock in the same region.
  - They occur because the surrounding rock and fault lines are shifting, adjusting, and settling down into a new stable position after the massive energy release.
  - They can continue happening for days, weeks, or even months, gradually decreasing in frequency and size.

By definition, the **mainshock** is the single largest earthquake in a given seismic sequence. Therefore, a foreshock cannot be larger than the mainshock because any earthquake that surpasses all previous magnitudes is automatically classified as the mainshock.

eto po yung nagiging problem po namin sir, in theory po kasi hindi po talaga pwedeng maging mas malaki yung foreshock kesa sa mainshock.

Recorded mainshock magnitude	Foreshocks in window	Largest foreshock magnitude
6.5	3	6.7
5.7	4	6.7
5	3	5.1
5.1	3	5.6
5.2	3	5.2
5.2	14	7.3
5.2	11	5.3
5	15	7.3
5.5	8	5.3
5	3	5.5
6.1	3	5.1
5.3	4	6.1

Bali ito pong Maeda's Method ganito po yung in theory niya po.
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3.1.1 Foreshock Identification Procedure

The method used to search for parameters of foreshocks that show high forecast performance consists of the following four steps:

1. Elimination of small aftershocks from the original data. Distance L (km) and the time interval t_a (days) between a mainshock and an aftershock depend on the magnitude of a previous earthquake (M_{pre}) as

$$\log_{10} L \le 0.5M_{\text{pre}} - 1.8, \quad (1)$$

$$\log_{10}(t_a + 0.3) \le \{0.17 + 0.85(M_{\text{pre}} - 4.0)\}/1.3, \quad (2)$$

where Equations 1 and 2 are basing on empirical laws proposed by Utsu (1961) and Utsu (1970), respectively, to identify aftershocks. For example, when M_{pre} is 8, 7, and 6, L is 160, 50, and 16 km, respectively, and t_a is 557, 123, and 27 days, respectively. To remove only small earthquakes, we introduced the following condition:

$$M_a < M_{\text{pre}} - M_d; \quad M_d = 1.0, \quad (3)$$

where M_a is the magnitude of an aftershock, and M_d represents the magnitude difference between M_{pre} and M_a. The reason for including Equation 3 is that earthquake swarm activity with small magnitude differences may be connected to a subsequent larger event. Note that although it has been reported that small aftershock data are missing in catalogs due to overlapping seismic waveform records for about one day immediately after a large earthquake (Zhuang et al., 2017), Maeda's method can avoid the problem of short-term aftershock incompleteness by not dealing with small earthquakes.

2. Selection of foreshock (precursory swarm activity) candidates, defined as the number N_f of earthquakes with magnitude ≥ M_{f0} (the foreshock magnitude threshold) occurring in a segment with size $D°$ × $D°$ (latitude × longitude) during a period of T_f days. Grid points at the centers of segments in the evaluation area are spaced at intervals of $D°/2$ (red crosses in Figures S1a and S5a). The alarm area is the same segment with size $D°$ × $D°$. The N_f-th foreshock candidate in each sequence is called an alarm earthquake in this paper

3. Selection of the alarm period. The alarm period T_a is the number of days after an alarming earthquake is used to forecast a target earthquake

4. The values of D, M_{f0}, T_f, N_f, T_a, and M_{m0} (the mainshock magnitude threshold) that result in a high forecast performance are determined by the grid-search method. See Section 3.1.2 for forecast efficiency

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Earthquakes with $M \ge M_{m0}$, excluding aftershocks (earthquakes that satisfy Equations 1 and 2), are defined as target earthquakes. We applied Maeda's method to the real catalog to obtain the optimum values of parameters D, M_{f0}, T_f, N_f, T_a, and M_{m0} in each study region. In this regard, for optimum values of M_{f0} and M_{m0}, and for optimum values of D in the areas along the Japan trench, we used the values determined by Maeda and Hirose (2016) in order to save calculation cost. To determine optimum values for D in central Honshu and the Izu Islands, we tested three possible values (Table S1). We estimated values for the other parameters by the grid search method from 900 combinations because it is sensitive to earthquake catalog: T_f = 1, 2, …, 10 days; N_f = 2, 3, …, 10 events; and T_a = 1, 2, …, 10 days. The optimum parameter values and forecast results (see Section 3.1.2) in each region in this study (Table 2) are different from those in the previous studies (Maeda, 1993, 1996; Maeda & Hirose, 2016) because of differences in the data precision flags, evaluation regions (this study considered only specific areas along the Japan trench), and analysis periods.